

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3445U20-1



MONDAY, 10 JUNE 2024 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life
FOUNDATION TIER

1 hour 30 minutes

ADDITIONAL MATERIALS

- Separate Resource Folder
- Calculator
- Pencil
- Ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	8	
	2.	5	
	3.	6	
	4.	10	
	5.	5	
	6.	3	
	7.	4	
	8.	9	
Section B	9.	6	
	10.	19	
	Total	75	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 8(b) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate resource folder to answer questions 9 and 10.

A Periodic Table is printed on page 28.



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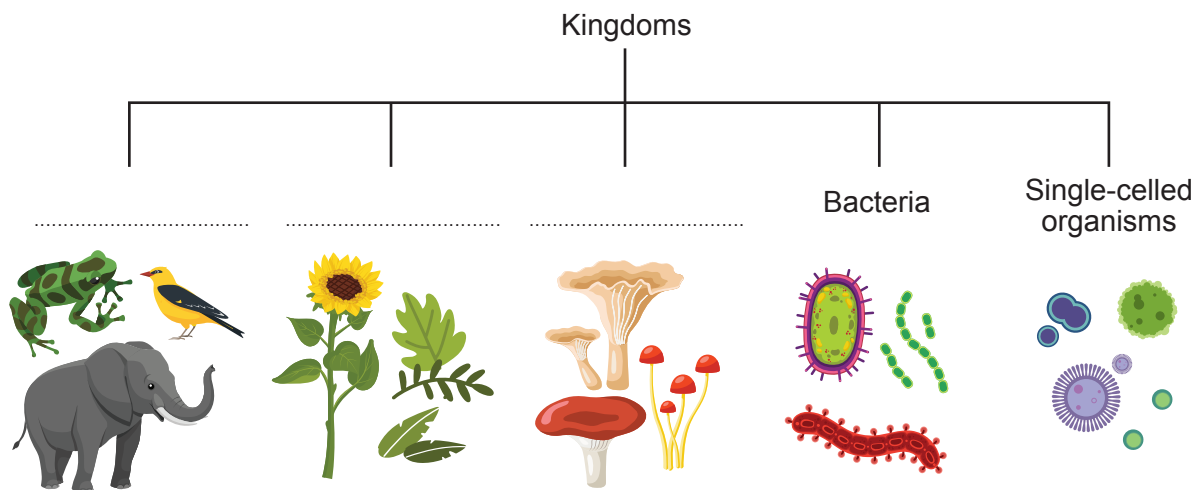
Section AAnswer **all** questions.

1. Living things are classified using a system developed by Carl Linnaeus.

(a) The first division of living things is to place them into a kingdom. There are 5 kingdoms.

Complete the labelling of the kingdoms on the diagram below.

[3]



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(b) Linnaeus’ system further ranked living things to a species level. This is illustrated below. The levels are not in the correct order.

A	B	C	D	E	F	G
species	order	class	phylum	genus	kingdom	family

Place the letters representing the levels in the correct order in the column below. Some have been done for you.

[2]

F
.....
C
B
.....
E
.....



(c) The classification of a brown bear is shown below. Its scientific name is *Ursus arctos*.

Kingdom	Animalia
	Chordata
Class	Mammalia
Order	Carnivora
	Ursidae
Genus	<i>Ursus</i>
	<i>arctos</i>



A wolf has the scientific name *Canis lupus*. Complete the classification table for a wolf. [3]

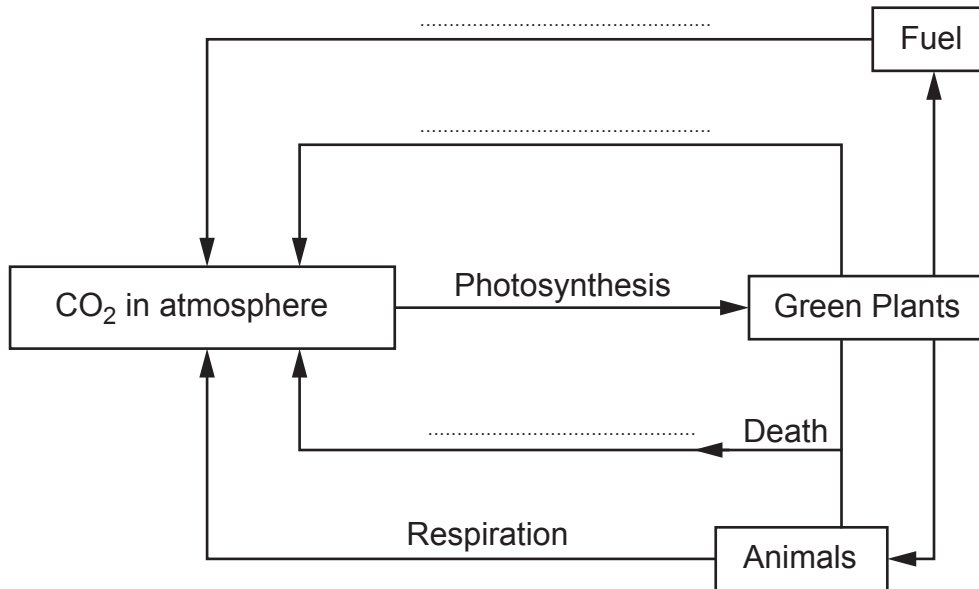
Kingdom	Animalia
	
Class	
Order	Carnivora
	Canidae
Genus	
	<i>lupus</i>

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2. (a) Label the carbon cycle diagram below. Use words from the box. [3]

dissolving evaporating decay respiration combustion



(b) Complete the following sentences by underlining the correct word in the brackets. [2]

- (i) When plant and animal waste decomposes, (**sulfates** / **nitrates** / **oxides**) are added to the soil.
- (ii) Plants use (**sulfates** / **nitrates** / **oxides**) in the soil to grow.



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3. Indicator species may be used in an investigation to monitor water pollution at three different locations along a stream.

- (a) The method for an investigation to monitor water pollution in a stream is described below. However, some of the stages are missing.

Complete the method by writing in the missing steps.

[2]

1. Take a tray and a net to the first location.
2.
3. Study the organisms in the tray.
4.
5. Pour the invertebrates and the water gently back into the stream.
6. Repeat steps 1–5 at the second and third locations.



(b) The following chart is available to you.

Pollution level

Indicator species

A Clean water



Stonefly nymph



Mayfly nymph

B Some pollution



Freshwater shrimp



Caddis fly larva

C Moderate pollution



Water louse



Bloodworm

D High pollution



Sludgeworm



Rat-tailed maggot

E Very High pollution – no life

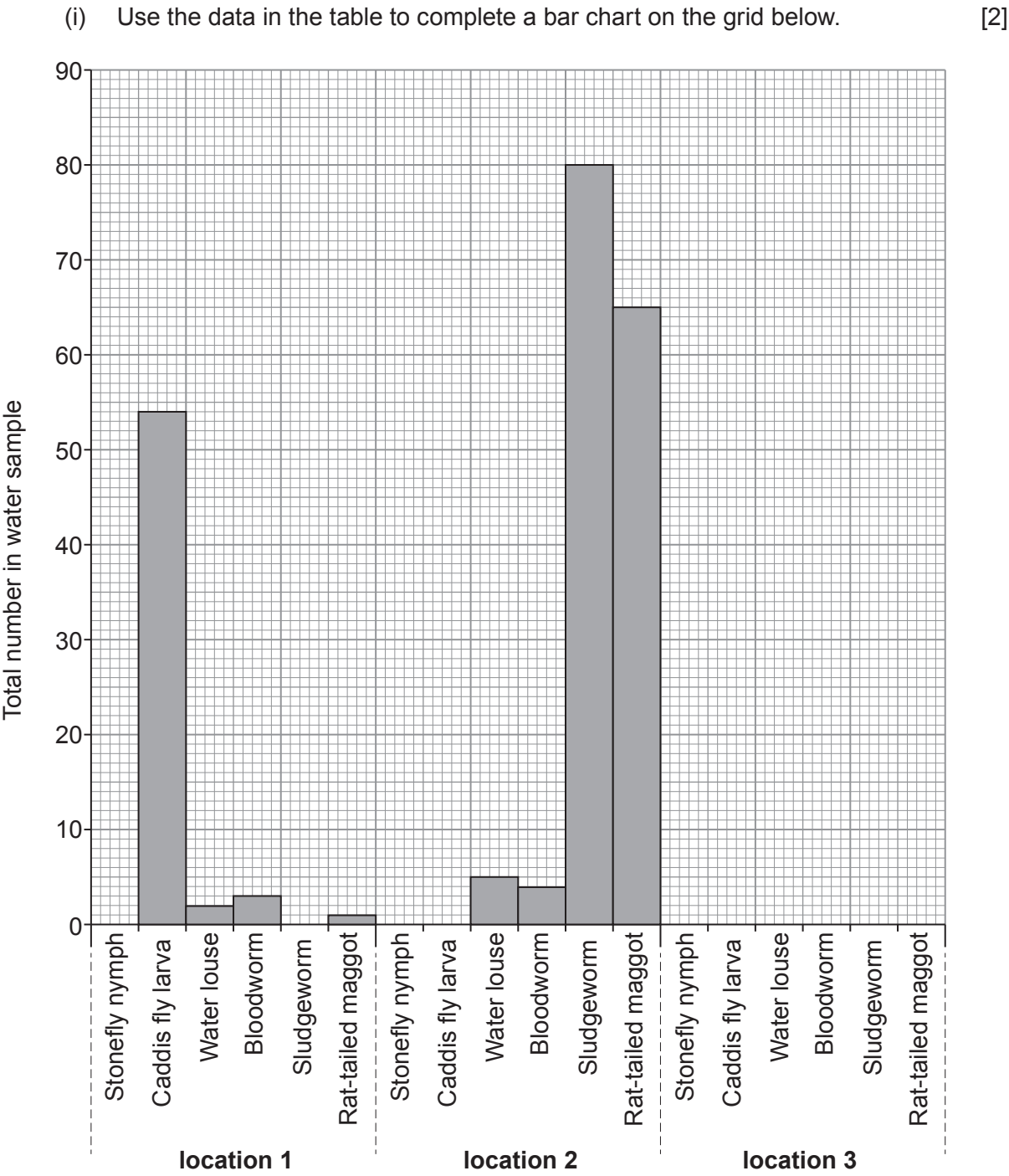


Results from the water sampling at each location are given in the table below.

Species	Total number in water sample		
	location 1	location 2	location 3
Stonefly nymph	0	0	31
Caddis fly larva	54	0	4
Water louse	2	5	3
Bloodworm	3	4	2
Sludgeworm	0	80	1
Rat-tailed maggot	1	65	1



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(ii) State which location is the cleanest. [1]

.....

(iii) Give **one** reason for your answer. [1]

.....
.....
.....



4. (a) Theories about the Universe have changed over time.

Tick (✓) the boxes next to the names of **two** of these theories.

[2]

Liquid State

☐

Steady State

☐

Loud Bang

☐

Big Bang

☐

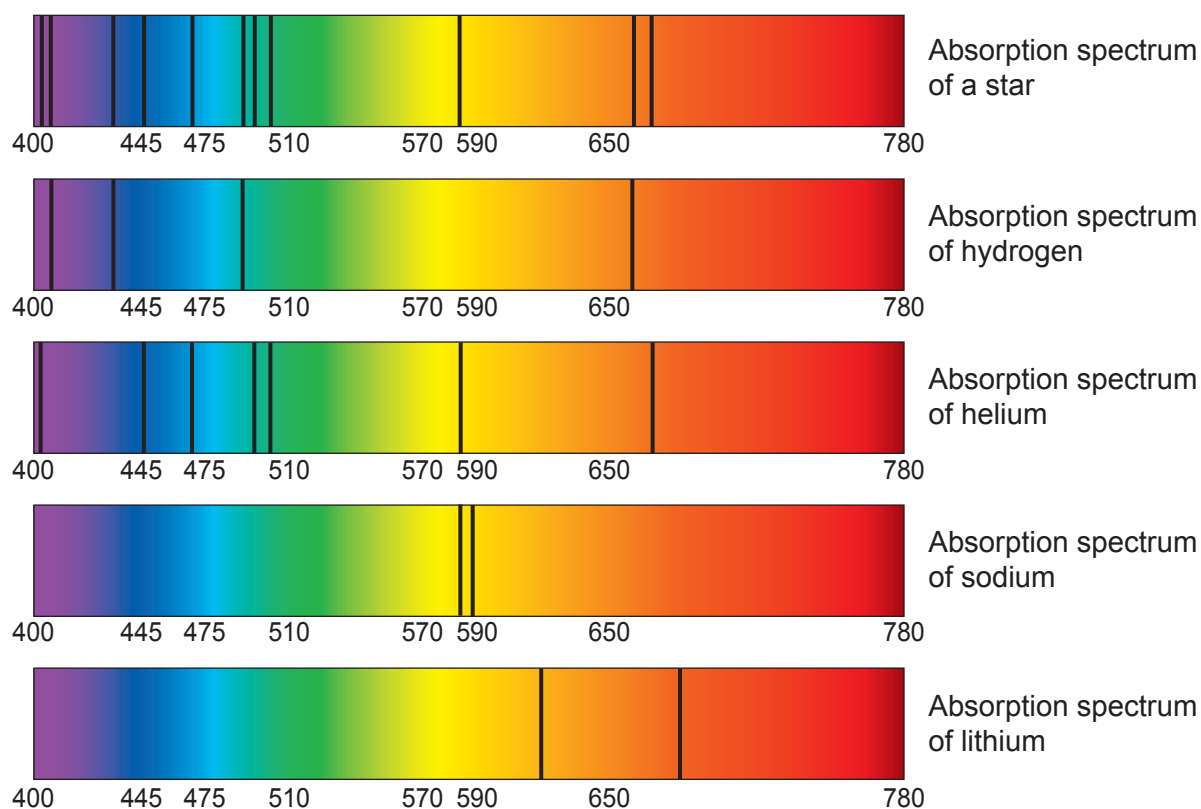
- (b) Images of the Universe are taken using electromagnetic waves.

Name **two** types of electromagnetic wave.

[2]

..... and

- (c) The diagram shows a spectrum from a star and four elements.



Complete the table by placing a tick (✓) if the element is present in the star. Hydrogen has been done for you.

[1]

Element	Element is present (✓)
hydrogen	✓
helium	
sodium	
lithium	



(d) The table gives information about the planets in our solar system.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
density (kg/m^3)	5427	5243	5514	3933	1326	687	1271	1638
mean distance from the Sun (million km)	58	108	150	228	779	1434	2873	4495
mean speed of orbit (km/s)	47.4	35.0	29.8	24.1	13.1	9.7	6.8	5.4
mean temperature ($^{\circ}\text{C}$)	167	464	15	-65	-110	-140	-195	-200
number of moons	0	0	1	2	79	82	27	14

Use the information in the table to answer the questions below.

(i) Name the coldest **rocky** planet. [1]

(ii) State how many **rocky** planets have moons. [1]

(iii) Name the gas giant that orbits the Sun with the greatest **speed**. [1]

(iv) Eris is a dwarf planet. Its orbit is 10 180 million km away from the Sun.
Shaun says it is found in the asteroid belt between the orbits of Mars and Jupiter.
Explain whether you agree with Shaun. [1]

.....

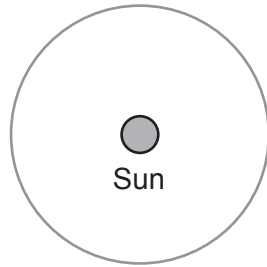
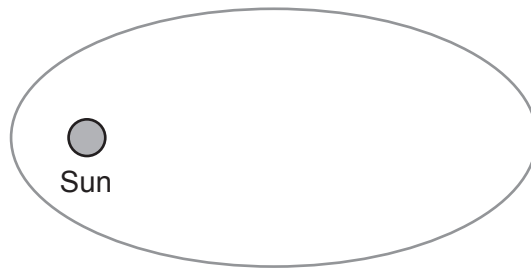
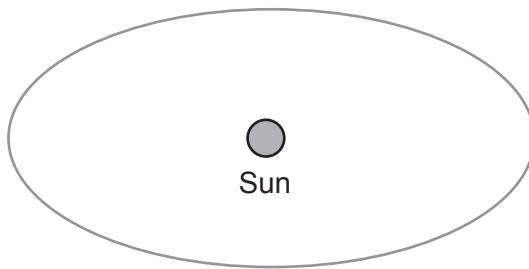
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(e) Tick (✓) the box next to the shape of the orbit of a comet around the Sun.

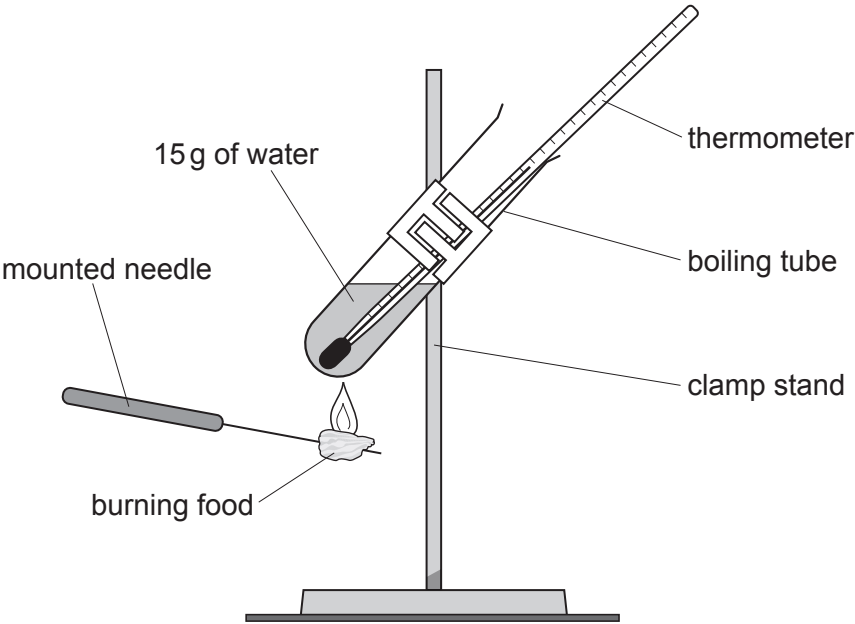
[1]

☐☐☐

10



5. Some students investigated the energy content of food.



Their results are shown in the table below.

Food	Mass of food burnt (g)	Initial water temperature (°C)	Final water temperature (°C)	Temperature increase (°C)
bread	2.5	18	23	5
cheese	4.0	17	31	14
cornflakes	3.0	17	29	12
oatie biscuit	5.0	18	25	7

(a) Use the equation

energy released = 15 × 4.2 × temperature increase

to calculate the energy released by the bread.

[2]

energy released = J



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- (b) The students were asked to use their results to calculate the energy provided by a snack consisting of 15 g of oatie biscuits and 20 g of cheese. Complete the tables below to work out their answer. [3]

Energy from 5 g of oatie biscuits (J)	Energy from 15 g of oatie biscuits (J)
441	

Energy from 4 g of cheese (J)	Energy from 20 g of cheese (J)
882	

Space for working

total energy from snack = J

5



6. All drugs can have side effects. New drugs undergo testing before they can be prescribed by doctors.

The following terms are used in drug development studies.

Join each term to its description with a straight line.

[3]

Term	Description
Blind study	an inactive substance made to look like the real drug
Double-blind study	when the patient and the researcher do not know if they are using the real drug
Placebo	when only the researcher does not know if they are using the real drug
	when the patient does not know if they are given the real drug

3



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7. Some of the methods used in the treatment and diagnosis of disease are included in the table below.

Place **one** tick (✓) in each row underneath the method used by each treatment or diagnosis.

[4]

Treatment or diagnosis	Uses high intensity light waves	Uses short wavelength electromagnetic waves	Uses high frequency sound waves	Uses strong magnetic fields	Uses drugs
chemotherapy					
ultrasound					
MRI scan					
CAT scan					

4



8. Vaccines are used to protect against disease.

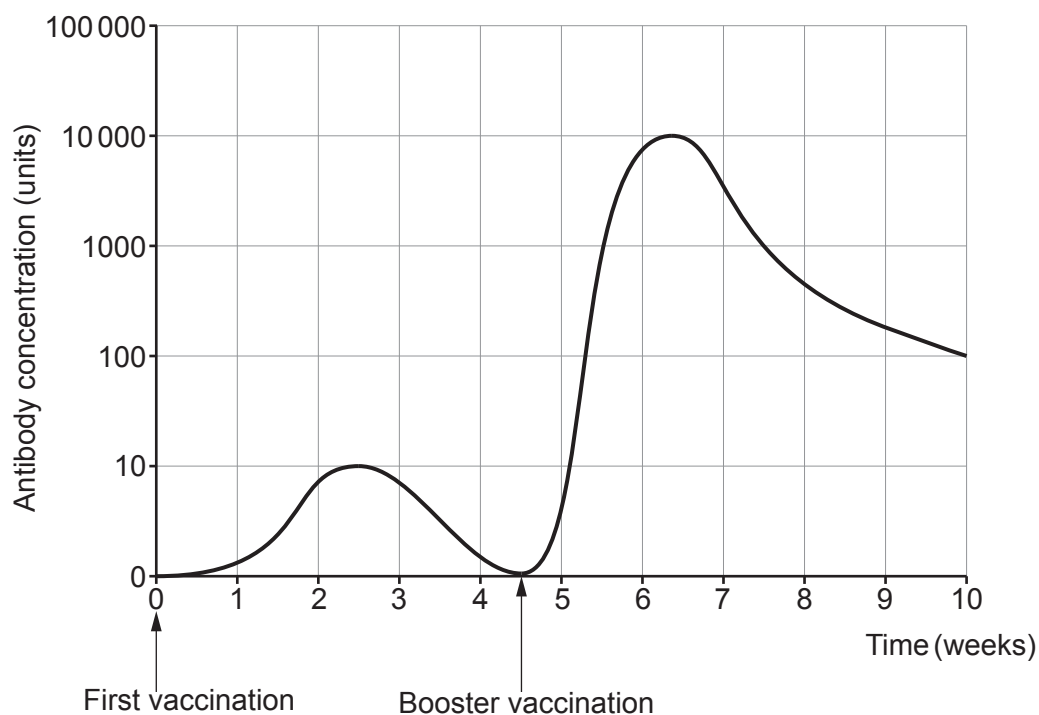
(a) Complete the following sentences about how the body reacts after infection or vaccination by underlining the correct term in the brackets.

(i) Vaccines contain molecules called (**antibodies** / **antigens** / **memory cells**). [1]

(ii) White blood cells respond to these molecules and produce (**antibodies** / **antigens** / **pathogens**). [1]

(iii) (**Antibiotics** / **antigens** / **memory cells**) are produced following infection or a vaccination. [1]

(b) The graph below shows the immune response over a 10-week period when a vaccine was given as two doses.



Describe the changes in antibody concentration over 10 weeks using data from the graph. [6 QER]

.....

.....

.....

.....



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9



Section B

Refer to the separate Resource Folder to answer questions 9 and 10.

9. Jake’s heart rate, breathing rate and airflow were monitored during a training session. Use the information in **Table 2** to answer the following questions.
- (a) State the value of the resting heart rate. [1]
heart rate = bpm
- (b) State the time at which the exercise stopped. [1]
time = min
- (c) Calculate the time the exercise lasted. [1]
time = min
- (d) (i) Calculate the drop in heart rate between 14 and 15 minutes. [1]
drop in heart rate = bpm
- (ii) Assuming that the heart rate drops at the same rate, calculate the heart rate after 16 minutes. [1]
heart rate after 16 minutes = bpm
- (e) Estimate the breathing rate after 16 minutes. [1]
breathing rate = breaths per minute

6



10. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct. [2]

.....

.....

.....

.....

- (b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula. [2]

.....

.....

- (c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

- (i) Calculate Malcolm's BMI. [2]

BMI =

- (ii) Calculate Malcolm's maximum heart rate. [1]

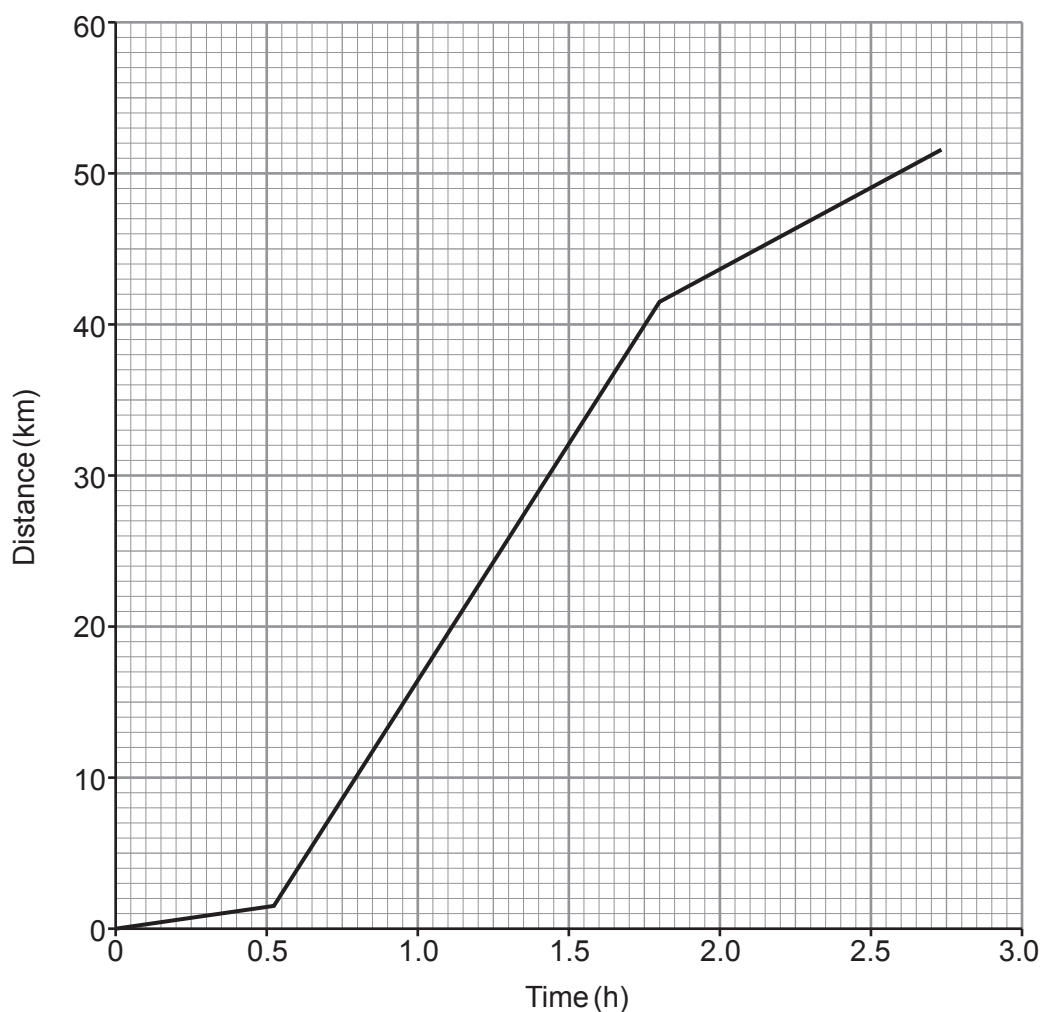
maximum heart rate = bpm



- (d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3 and 4** and on page 4 to determine the range in heart rate she should aim for. [1]

heart rate range is from bpm to bpm.

- (e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.



Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon:

Reason 1:

Reason 2:

- (f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = km/h

- (g) When a triathlete mounts their bike they are travelling at 1.5 m/s. Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = m/s²

- (h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

.....

.....

.....

.....

END OF PAPER



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	Group
1	1
2	2
3	3
4	4
5	5
6	6
7	7
0	0

1 H Hydrogen 1 4 He Helium 2																																											
7 Li Lithium 3		9 Be Beryllium 4		23 Na Sodium 11		24 Mg Magnesium 12		39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54									
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86									
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																																							

Key

Key

- relative atomic mass

Ar
Symbol
Name
Z
atomic number